CELLULAR NETWORK DESIGN

Pombhurna, Chandrapur District, Maharashtra

4G LTE Coverage Design & MIMO Channel Modelling

ASSIGNMENT 1 & 2 — WIRELESS COMMUNICATION

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1. Site Analysis Report – Pombhurna

1.1 Town Overview and Geography

Pombhurna is a nagar panchayat and small town located in Chandrapur district, Maharashtra, falling under the Rajura tehsil of the Vidarbha region. The town lies approximately 80 km south-east of Chandrapur city and has a population of approximately 20,000–25,000 (Census 2011). The settlement occupies a roughly 4 km × 3 km footprint, giving an estimated planning area of 12 km².

The terrain is largely flat to gently undulating, characteristic of the eastern Deccan Plateau, at an elevation of approximately 230–250 m above mean sea level. The surrounding land is predominantly rain-fed agricultural (cotton, soybean), interspersed with patches of scrub forest and the Penganga River system in the south-western quadrant.

1.2 Key Features: Roads, Rivers, Markets, and Population Density

Roads and Highway Network

National Highway NH-930 (formerly SH-263) is the principal artery passing through Pombhurna, connecting it northward to Wani and southward to Rajura. The highway carries heavy goods vehicles, passenger buses, and private traffic. The stretch through the town core is approximately 1.5 km long and constitutes the highest-priority coverage corridor. Secondary roads branch from NH-930 into residential mohallas and toward peripheral hamlets, forming a roughly linear settlement pattern aligned with the highway axis.

River and Water Bodies

The Penganga River — a major right-bank tributary of the Godavari — lies approximately 3–5 km south-west of the town centre. The river valley creates a broad, low-lying corridor. While the floodplain itself has minimal habitation, the valley geometry can induce atmospheric ducting and ground-reflection multipath effects relevant to cell fringe performance. The eastern boundary of the town is marked by a minor seasonal stream.

Market and Commercial Core

The primary commercial zone — locally referred to as the main bazaar — is concentrated along NH-930 for approximately 500 m on either side of the central chowk. This area hosts retail shops, a grain mandi, cooperative bank branches, government offices, and a weekly market (haat). User density peaks during morning (9 AM–12 PM) and evening (5–8 PM) windows. This zone demands the highest cellular capacity and is designated a critical coverage priority.

Population and Building Density

The central residential mohallas flanking the market feature 2–3 storey load-bearing masonry structures with moderate indoor penetration loss (10–15 dB). North and south of the highway are medium-density single-storey residential areas with open compounds. The eastern and western fringes transition to widely spaced agricultural homesteads with near-zero indoor penetration requirements. The south-west sector toward Penganga is the least populated zone.

Market / NH-930	Dense commercial	2–3 floors masonry	Very High	Critical
Central Residential	Urban residential	1–2 floors	High	High
N & S Residential	Semi-urban	1 floor	Medium	High
East Agricultural Fringe	Peri-urban	Scattered homesteads	Low	Medium
West / Penganga	Rural / river buffer	Very sparse	Very Low	Low

Figure 1.1 — Site Analysis Map: Pombhurna, Chandrapur District, Maharashtra

Left: Satellite imagery with zone overlays. Right: Schematic site analysis map with legend.

1.3 Design Considerations from Site Analysis

- **NH-930 corridor demands high-capacity cells with smaller radii and potentially a dedicated microcell to handle peak vehicular and pedestrian traffic.**
- **The Penganga river corridor (west/south-west) can be served with large-radius cells — sparse population reduces capacity requirements and justifies lower CAPEX in that zone.**
- **Low-rise building stock (predominantly 1–2 floors) results in moderate indoor penetration loss of 10 dB, as assumed in the link budget.**
- **Flat terrain favours macro-cell deployment but cannot guarantee LOS in built-up zones — the conservative UMa-NLOS path loss model is mandated.**
- **The market area must be treated as a capacity bottleneck; a heterogeneous network (HetNet)**

architecture with an umbrella macro and market microcell is the recommended approach.

- **Candidate tower sites include water tower structures, the tallest buildings along NH-930 (3–4 floors), and monopoles on government land near the bus stand.**

2. PART 1 – 4G LTE Coverage Network Design

2.1 Assumptions

The following assumptions govern the network design. Each is justified by standard 4G deployment practice in India and by the physical characteristics of Pombhurna.

- **The 4G system operates on LTE-FDD Band 3 at 1800 MHz — the most widely deployed 4G band in India (Airtel, Jio, Vi).**
- **Base station (eNB) transmit power: 46 dBm (40 W) — standard macro eNB in a semi-urban environment.**
- **Three-sector macro antennas with 17 dBi gain and 120° azimuth beamwidth are used per site.**
- **BS antenna height hBS = 25 m, consistent with 3GPP TR 38.901 Table 7.4.1-1 UMa parameters.**
- **UE height hUT = 1.5 m (pedestrian / in-vehicle standard assumption).**
- **Cable and feeder losses at the BS: 3 dB (tower-top installation).**
- **UE receiver noise figure: 7 dB — representative of a mid-range 4G smartphone.**
- **System bandwidth: 10 MHz (50 resource blocks, LTE FDD).**
- **Minimum cell-edge SNR: 3 dB — corresponding to QPSK rate 1/3, the most robust LTE MCS, adequate for voice and low-rate data.**
- **Log-normal shadowing margin: 8 dB, targeting 90% coverage probability in a semi-urban environment.**
- **Indoor penetration loss: 10 dB — applicable to the low-rise masonry buildings prevalent in Pombhurna.**
- **Frequency reuse cluster size N = 7 (D/R = 4.583) for the macro layer.**
- **Hexagonal cell geometry is used for coverage area estimation.**

2.2 System Parameters Summary

Carrier Frequency	fc	1800	MHz	LTE Band 3
BS Transmit Power	Pt	46	dBm	= 40 W, macro eNB
BS Antenna Gain	GBS	17	dBi	3-sector directional
Cable / Feeder Loss	Lf	3	dB	BS side
UE Antenna Gain	GUE	0	dBi	Omnidirectional
Body Loss	Lb	3	dB	Human body absorption
UE Noise Figure	NF	7	dB	Typical 4G UE
System Bandwidth	BW	10	MHz	50 RBs, LTE FDD

Min Required SNR	SNRmin	3	dB	QPSK 1/3 — cell edge
Shadowing Margin	Msf	8	dB	90% coverage probability
Penetration Loss	Lp	10	dB	Low-rise masonry
BS Antenna Height	hBS	25	m	3GPP UMa default
UE Height	hUT	1.5	m	Pedestrian standard

2.3 Maximum Allowable Path Loss (L_max)

The Maximum Allowable Path Loss (MAPL), denoted L_max, is the maximum signal attenuation the link budget can tolerate while maintaining acceptable service quality at the cell edge. It is formally defined as:

$$L_max = P_t,dBm - P_r,min,dBm$$

We first compute the effective isotropic radiated power (EIRP) at the BS:

$$EIRP = P_t + G_BS - L_feeder = 46 + 17 - 3 = 60 \text{ dBm}$$

Next, the minimum receiver sensitivity P_r,min is derived from the thermal noise floor:

$$\text{Thermal Noise} = -174 + 10 \cdot \log10(10 \times 10^6) = -174 + 70 = -104 \text{ dBm}$$

$$\text{Noise Floor} = -104 + NF = -104 + 7 = -97 \text{ dBm}$$

$$P_r,min = \text{Noise Floor} + SNR_min = -97 + 3 = -94 \text{ dBm}$$

Combining EIRP, receiver sensitivity, and all applicable losses and margins:

$$L_max = EIRP + G_UE - L_body - M_sf - L_pen - P_r,min$$

$$L_max = 60 + 0 - 3 - 8 - 10 - (-94) = 133.00 \text{ dB}$$

EIRP (Pt + GBS – Lf)	+60 dBm	+
UE Antenna Gain GUE	0 dBi	+
Body Loss Lb	3 dB	–
Shadowing Margin Msf	8 dB	–
Indoor Penetration Loss Lp	10 dB	–
Minimum Rx Sensitivity Pr,min	–94 dBm	–
Maximum Allowable Path Loss Lmax	133.00 dB	=

2.4 Path Loss Model Selection and Justification

3GPP TR 38.901 V18.0.0 (Table 7.4.1-1) specifies path loss models for several propagation scenarios: Indoor Hotspot (InH), Urban Micro (UMi), Urban Macro (UMa), and Rural Macro (RMa). The correct scenario must be selected based on the deployment geometry and environment.

Selected Model: UMa-NLOS (Urban Macro, Non-Line-of-Sight)

Justification for choosing UMa over other models:

- **Inter-site distances of 1–2 km are planned, which is the characteristic scale of Urban Macro deployments. RMa applies to ISD > 5 km; UMi to ISD < 200 m.**
- **The BS antenna at hBS = 25 m is above the average rooftop (~8–10 m), satisfying the UMa condition hBS > h_rooftop.**
- **Operating frequency 1.8 GHz is within the UMa model validity range of 0.5–100 GHz.**

Justification for choosing NLOS over LOS:

- **LOS cannot be guaranteed across the entire cell coverage area in the market and residential zones — buildings along NH-930 create frequent blockage.**
- **Using NLOS provides a conservative (pessimistic) estimate of Rmax, ensuring the design does not overestimate coverage range.**
- **A higher-attenuation model also accounts for realistic clutter in a semi-urban environment.**

UMa-NLOS Path Loss Formula (3GPP TR 38.901, Table 7.4.1-1):

$$PL_UMa\text{-}NLOS = 13.54 + 39.08 \cdot \log10(d3D) + 20 \cdot \log10(fc) - 0.6 \cdot (hUT - 1.5) \ [dB]$$

Where:

- **$d3D = \sqrt{(d2D^2 + (hBS - hUT)^2)}$ [3D distance in metres]**
- **fc = carrier frequency in GHz = 1.8**
- **hBS = 25 m, hUT = 1.5 m**
- **Validity range: 10 m ≤ d2D ≤ 5000 m**

Note: The model uses max(PL_LOS, PL_NLOS) to ensure the conservative bound is always applied. For d2D > ~500 m at 1.8 GHz with hBS = 25 m, the NLOS term dominates.

2.5 Finding R_max from L_max

R_max is the maximum permissible cell radius — the distance at which the UMa-NLOS path loss exactly equals L_max = 133.00 dB. It is found by inverting the path loss model:

$$\text{Solve: } PL_UMa\text{-}NLOS(R_max) = L_max = 133.00 \text{ dB}$$

A binary search is performed numerically (see Python code, Section 4). The solution converges to:

$$R_max = 843 \text{ m} \approx 0.843 \text{ km}$$

Verification at Rmax = 843 m:

$$d3D = \sqrt{(843^2 + (25 - 1.5)^2)} = \sqrt{(710649 + 552)} = 843.3 \text{ m}$$

$$PL = 13.54 + 39.08 \cdot \log10(843.3) + 20 \cdot \log10(1.8) - 0$$

PL = 13.54 + 114.35 + 5.11 = 133.00 dB ✓

100	101.4	97.26	+35.74	Well inside range
200	201.4	108.69	+24.31	Inside range
500	500.6	124.14	+8.86	Inside range
800	800.3	132.11	+0.89	Just inside
843	843.3	133.00	0.00	← R_max
1000	1000.3	135.89	–2.89	Outside limit
1200	1200.2	138.98	–5.98	Outside limit
1500	1500.2	142.77	–9.77	Outside limit
2000	2000.1	147.65	–14.65	Far outside

2.6 Initial Network Design – D/R Ratio

For a hexagonal cellular layout, the co-channel reuse distance D and cell radius R are related to the cluster size N by:

$$D/R = \sqrt{(3N)}$$

A cluster size of N = 7 is selected for the macro layer. This is the most commonly employed reuse factor in practical LTE deployments globally, offering a sound balance between spectral efficiency and co-channel interference protection. The resulting D/R ratio and separation are:

$$D/R = \sqrt{(3 \times 7)} = \sqrt{21} = 4.583$$

With N = 7, each cell uses 1/7 of the total available spectrum. The co-channel C/I under NLOS conditions exceeds 18 dB, which is sufficient for LTE QPSK reception.

Initial trial cell radii, based on user density and coverage priority:

Cell 1	Market / NH-930 corridor	0.80	High density — microcell scale
Cell 2	Main residential core	1.20	Medium density urban
Cell 3	North residential	1.20	Medium density urban
Cell 4	South residential	1.20	Medium density urban
Cell 5	East agricultural fringe	1.80	Low density peri-urban
Cell 6	West fringe / Penganga	1.80	Very low density rural
Cell 7	South-west rural buffer	2.00	Rural sparse settlement

2.7 Iteration 1 – Coverage Verification (Several Cells Fail)

Each cell is checked against R_max = 0.843 km. A cell PASSES if and only if the UMa-NLOS path loss at its assigned radius R is ≤ L_max = 133.00 dB.

1	Market / Commercial	0.80	132.11	133.00	YES	✓ PASS
2	Residential Core	1.20	138.98	133.00	NO	✗ FAIL
3	North Residential	1.20	138.98	133.00	NO	✗ FAIL
4	South Residential	1.20	138.98	133.00	NO	✗ FAIL
5	East Fringe	1.80	145.86	133.00	NO	✗ FAIL
6	West / Penganga	1.80	145.86	133.00	NO	✗ FAIL
7	Rural Buffer	2.00	147.65	133.00	NO	✗ FAIL

Observation: Cells 2 through 7 fail the coverage criterion. Their path loss at the trial radii exceeds L_max = 133.00 dB by between 5.98 dB and 14.65 dB. A redesign is required.

2.8 Iteration 2 – Redesigned Network (All Cells Pass)

Two corrective strategies are applied following the failed Iteration 1:

- Reduce all cell radii to R = 0.80 km (≤ R_max = 0.843 km). This requires more cells to cover the 12 km² town area. With hexagonal cell area = 2.598 × R² = 2.598 × 0.64 = 1.663 km², approximately 8 cells are needed for full coverage.
- For low-density rural fringe cells, an alternative is to accept a coverage gap (no subscribers) or use a taller 40 m mast, which shifts the LOS probability upward and allows a larger effective radius. Here, Strategy 1 is applied uniformly for consistency.

Redesigned Cell Plan — Iteration 2:

1	Market / NH-930	0.80	25	UMa-NLOS	132.11	133.00	✓
2	Residential Core	0.80	25	UMa-NLOS	132.11	133.00	✓
3	North Residential	0.80	25	UMa-NLOS	132.11	133.00	✓
4	South Residential	0.80	25	UMa-NLOS	132.11	133.00	✓
5	East Fringe	0.80	25	UMa-NLOS	132.11	133.00	✓
6	West Fringe	0.80	25	UMa-NLOS	132.11	133.00	✓
7	Rural Buffer N	0.80	25	UMa-NLOS	132.11	133.00	✓
8	Rural Buffer S	0.80	25	UMa-NLOS	132.11	133.00	✓

All 8 cells in the redesigned network satisfy R < R_max = 0.843 km. At R = 0.80 km, the path loss is 132.11 dB — 0.89 dB below L_max, confirming a positive link margin. The network is declared coverage-valid.

2.9 Microcell and Umbrella Cell Justification

Cell 1 — Market Microcell:

The NH-930 market zone presents uniquely high user density combined with a street-canyon propagation environment. A dedicated microcell (Cell 1) with reduced transmit power (Pt = 33 dBm, 2 W) and R = 0.80 km is deployed here to provide high spectral efficiency. The lower power reduces co-channel interference with adjacent macro cells.

Cell 2 — Umbrella Macro Cell:

Cell 2 (Residential Core macro) functions as the umbrella cell over the microcell layer. It operates with higher power and elevated antenna downtilt to avoid interference with Cell 1 while ensuring seamless handoff for vehicles travelling at 60–80 km/h along NH-930. The umbrella cell also fills coverage gaps at the micro-cell periphery during periods of antenna loading or temporary hardware faults.

Benefits of the HetNet Architecture:

- **Microcell Cell 1 delivers spectral efficiency > 3 bps/Hz in the market zone.**
- **Umbrella cell supports continuous mobility management and inter-layer handover via X2 interface.**
- **Self-Organising Network (SON) algorithms manage transmit power and tilt dynamically to minimise inter-layer interference.**

3. PART 2 – Small Scale Fading: 3GPP TR 38.901 Section 7.5

3.1 Introduction to the Section 7.5 Procedure

Section 7.5 of 3GPP TR 38.901 V18.0.0 specifies a geometry-based stochastic channel model (GBSM) procedure for generating spatially consistent MIMO channel matrices suitable for link-level and system-level simulation. The procedure decomposes multipath propagation into N clusters, each representing a group of scatterers. Each cluster contains M rays, parameterised by delay, power, angles of departure and arrival (in both azimuth and elevation), and complex polarisation phases.

The following implementation uses UMa-NLOS parameters from Table 7.5-6, applied to the main residential macro cell (Cell 2) of the Pombhurna network at 1.8 GHz. A 4T×2R MIMO configuration is modelled, representing a standard LTE downlink with four BS antennas and two UE antennas.

3.2 Step 1 – Scenario and Array Parameters

Scenario	UMa-NLOS	Table 7.5-6, 3GPP TR 38.901
Carrier frequency fc	1.8 GHz	LTE Band 3
Wavelength λ	16.67 cm	λ = c / fc
BS Array (TX)	4 elements, ULA	λ/2 = 8.33 cm spacing
UE Array (RX)	2 elements, ULA	λ/2 = 8.33 cm spacing
MIMO Configuration	4T2R	4 transmit, 2 receive
Clusters N_cl	20	UMa-NLOS, Table 7.5-6
Rays per cluster M	20	Table 7.5-1
Array pattern	ULA omnidirectional	Simplified for demonstration

3.3 Step 2 – Large Scale Parameter Generation (Table 7.5-6)

Large scale parameters (LSPs) are drawn from log-normal distributions with parameters taken from 3GPP TR 38.901 Table 7.5-6 for the UMa-NLOS scenario. In a full implementation, the LSPs are spatially correlated using the cross-correlation matrix via Cholesky decomposition; here, each parameter is sampled independently for clarity.

Delay Spread (DS)	–6.955	0.66	–6.627	235.96 ns
AoA Spread (ASA)	1.81°	0.20	1.783	60.58°
AoD Spread (ASD)	1.59°	0.28	1.771	59.07°
ZoA Spread (ZSA)	0.88°	0.31	1.352	22.50°
ZoD Spread (ZSD)	0.50°	0.35	0.419	2.62°

3.4 Step 3 – Cluster Delays (Equation 7.5-1)

Cluster delays are generated using Eq. 7.5-1 of 3GPP TR 38.901:

$$\tau_n' = -r_\tau \cdot DS \cdot \ln(X_n), \quad X_n \sim \text{Uniform}(0, 1)$$

$$\tau_n = \text{sort}(\tau_n') - \min(\tau_n') \quad \text{[normalised, starts at 0]}$$

Where $r_\tau = 2.3$ is the delay distribution proportionality factor for UMa-NLOS (Table 7.5-6).

1	0.000	Reference delay (normalised to zero)
2	61.39	
3	82.95	
4	114.67	
5	170.76	
…	…	20 clusters in total

3.5 Step 4 – Cluster Powers (Equation 7.5-4)

Cluster powers are computed from Eq. 7.5-4:

$$P_n' = \exp(-\tau_n \cdot (r_\tau - 1) / (r_\tau \cdot DS)) \times 10^{(-Z_n / 10)}$$

$$P_n = P_n' / \Sigma P_n' \quad [\text{normalised}, \Sigma P_n = 1]$$

Where $Z_n \sim N(0, \zeta^2)$ with $\zeta = 3$ dB (per-cluster shadowing, Table 7.5-6). The three strongest clusters account for approximately 40% of total channel power, consistent with realistic multi-cluster propagation.

Total power check: $\Sigma P_n = 1.000$ ✓

3.6 Step 5 – Cluster Angles (Equations 7.5-6a/b)

Azimuth angle-of-arrival (AoA) cluster angles are generated per Eq. 7.5-6a:

$$\phi_{n,AOA} = (2 \cdot ASA / 1.4) \cdot \sqrt{(-\ln(P_n / \max(P_n)))} / C_\phi \times s_n + \Delta_n$$

Similarly for AoD, ZoA, and ZoD. Constants from Tables 7.5-2 and 7.5-3 for $N_{cl} = 20$:

- **$C_\phi = 1.289$ (azimuth angle scaling)**
- **$C_\theta = 1.178$ (elevation angle scaling)**

1	–54.47	54.51	88.42	82.83
2	–68.09	43.77	91.86	70.48
3	–12.35	7.11	90.58	90.97
4	–51.65	56.29	91.55	105.69
5	–71.61	–78.29	87.37	64.71
…	…	…	…	20 clusters total

3.7 Step 6 – Polarisation, Per-Ray Phases, and Channel Matrix H

For each cluster n and ray m, a 2×2 complex polarisation coupling matrix Ψ is formed using four independent initial phases $\Phi_{nm}^{\alpha\beta} \sim \text{Uniform}(-\pi, \pi)$ and the cross-polarisation power ratio (XPR) $\varkappa$:

$$\Psi_{nm} = \begin{bmatrix} \exp(j\Phi^{VV}) & (1/\sqrt{\varkappa}) \cdot \exp(j\Phi^{VH}) \\ (1/\sqrt{\varkappa}) \cdot \exp(j\Phi^{HV}) & \exp(j\Phi^{HH}) \end{bmatrix}$$

Where $\varkappa_{dB} \sim N(7.0, 3.0^2)$ dB per ray (Table 7.5-6, UMa-NLOS).

ULA array response vector for N elements with $\lambda/2$ spacing:

$$F(\phi) = [\, 1,\ e^{j \cdot \pi \cdot \sin(\phi)},\ e^{j \cdot 2\pi \cdot \sin(\phi)},\ e^{j \cdot 3\pi \cdot \sin(\phi)} \,]^T \quad \text{(4-element TX)}$$

The H matrix ($N_{UE} \times N_{BS} = 2 \times 4$) is assembled by summing over all clusters and rays:

$$H = \Sigma_n \sqrt{(P_n)} \cdot \Sigma_m [\, \Psi^{VV}_{nm} \cdot F_{RX}(\phi_{AOA,nm}) \otimes F_{TX}(\phi_{AOD,nm})^H \,] \,/\, \sqrt{M}$$

Computed H matrix (2×4, complex, UMa-NLOS at 1.8 GHz):

UE Antenna 1	–0.2738 – 0.6257j	–0.1626 + 1.0580j	+0.3385 – 0.3385j	–0.9464 + 0.0844j
UE Antenna 2	+0.9679 + 0.3161j	–0.4429 – 0.3375j	+0.2591 + 0.1095j	–0.0581 – 0.2167j

SVD Analysis:

$$H = U \cdot \Sigma \cdot V^H \quad \textbf{Singular values: } \sigma_1 = 1.9402, \ \sigma_2 = 0.6756$$

Matrix dimensions	2 × 4 (complex)	2 RX, 4 TX antennas
Rank	2 (full rank)	2 independent MIMO streams possible
Singular value σ_1	1.9402	Dominant spatial mode
Singular value σ_2	0.6756	Secondary spatial mode
Condition number σ_1/σ_2	2.87	Well-conditioned — good spatial separation
Frobenius norm $\|H\|_F$	2.0544	Total channel power
Estimated capacity (SNR=3dB)	1.82 bps/Hz	Shannon bound, equal power TX

3.8 Bonus: Steps 7–12 Discussion

Step 7 – Doppler Shift

For a UE travelling at velocity v along NH-930, the Doppler shift for each ray is given by:

$$f_{d,n} = (v / \lambda) \cdot \cos(\phi_{AOA,n} - \theta_v)$$

At v = 50 km/h = 13.9 m/s on the highway:

$$f_{d,max} = v / \lambda = 13.9 / 0.1667 = 83.3 \text{ Hz}$$

$$\textbf{Coherence time } T_c \approx 1 / (2 \cdot f_{d,max}) = 6.0 \text{ ms}$$

The coherence time of 6 ms is comparable to the LTE TTI of 1 ms, confirming that the channel varies on a sub-frame timescale for highway users and requires fast closed-loop channel estimation.

Step 8 – Time-Varying Phase

A time-dependent phase factor is multiplied onto each ray to model UE motion:

$$H(t) \mathrel{+}= \exp(j \cdot 2\pi \cdot f_{d,n} \cdot t) \cdot H_{nm} \quad \textbf{per ray}$$

Step 9 – Channel Impulse Response $h(t, \tau)$

$$h(t, \tau) = \Sigma_n \Sigma_m H_{nm}(t) \cdot \delta(\tau - \tau_n - \Delta\tau_{nm})$$

Each cluster contributes delta functions at its mean delay τ_n. The intra-cluster delay spread $\Delta\tau_{nm}$ accounts for sub-cluster ray offsets. The resulting wideband power delay profile determines the frequency selectivity of the channel.

Step 10 – Antenna Radiation Patterns

In a complete implementation, each element's radiation pattern $F(\theta, \phi)$ from 3GPP TR 38.901 Section 7.3 — including front-to-back ratio and vertical downtilt — modifies the effective signal at each cluster angle. For this simplified implementation, omnidirectional ULA patterns are assumed.

Step 11 – Large-Scale Path Loss and Shadowing Integration

The complete received signal model is: $y = \sqrt{PL_{total}} \cdot H \cdot x + n$, where PL_{total} incorporates the UMa-NLOS path loss (Section 2.4), log-normal shadowing, and the small-scale fading H from Step 6. This links Part 1 and Part 2 of this assignment into a unified system model.

Step 12 – Final Channel Matrix for Simulation

The complete H(t) time series forms the input to LTE link-level MIMO simulators. It can be used to evaluate capacity, BER under HARQ, and spatial multiplexing gain under realistic propagation conditions representative of Pombhurna's semi-urban environment. For system-level simulation, one H matrix per active UE is generated to study inter-cell interference and SINR distributions.

4. Python Code Listing

The complete Python code used for all calculations in this report is provided below. It is self-contained and reproducible. The seed is fixed (np.random.seed(42)) to ensure consistent outputs.

File: pombhurna_calculations.py

```
"""
Pombhurna 4G Network Design + 3GPP TR 38.901 Channel Modelling
Reference: 3GPP TR 38.901 V18.0.0 (2024-05)
"""
import numpy as np
import math

np.random.seed(42)  # Reproducible results

# ── PART 1: SYSTEM PARAMETERS ──────────────────────────────────────────
fc_MHz          = 1800      # LTE Band 3, MHz
fc_GHz          = 1.8
Pt_dBm          = 46        # BS transmit power, dBm
antenna_gain_BS   = 17        # dBi, 3-sector macro
cable_feeder_loss = 3         # dB
antenna_gain_UE   = 0         # dBi, omnidirectional
body_loss         = 3         # dB
```

```
noise_figure_UE   = 7          # dB
thermal_noise     = -174       # dBm/Hz at 300 K
bandwidth_Hz      = 10e6       # 10 MHz
SNR_min_dB        = 3          # dB, QPSK 1/3 at cell edge
shadow_margin     = 8          # dB, 90% coverage probability
penetration_loss  = 10         # dB, low-rise buildings

# Receiver sensitivity
N_dBm       = thermal_noise + 10 * math.log10(bandwidth_Hz)
noise_floor = N_dBm + noise_figure_UE
Pr_min_dBm  = noise_floor + SNR_min_dB   # = -94.00 dBm

# Maximum allowable path loss
EIRP = Pt_dBm + antenna_gain_BS - cable_feeder_loss   # = 60 dBm
Lmax = (EIRP + antenna_gain_UE - body_loss
        - shadow_margin - penetration_loss - Pr_min_dBm)
# Lmax = 133.00 dB

# ── UMa-NLOS MODEL (3GPP TR 38.901 Table 7.4.1-1) ──────────────
def path_loss_UMa_NLOS(d2D_m, fc_GHz, hBS=25, hUT=1.5):
    d3D     = math.sqrt(d2D_m**2 + (hBS - hUT)**2)
    PL_NLOS = (13.54 + 39.08*math.log10(d3D)
               + 20*math.log10(fc_GHz)
               - 0.6*(hUT - 1.5))
    d_BP    = 4 * hBS * hUT * fc_GHz * 1e9 / 3e8
    if d2D_m < d_BP:
        PL_LOS = 28.0 + 22*math.log10(d3D) + 20*math.log10(fc_GHz)
    else:
        PL_LOS = (28.0 + 40*math.log10(d3D) + 20*math.log10(fc_GHz)
                  - 9*math.log10(d_BP**2 + (hBS-hUT)**2))
    return max(PL_NLOS, PL_LOS)  # conservative bound

def find_rmax(Lmax_dB, fc_GHz, hBS=25, hUT=1.5):
    lo, hi = 10, 50000          # binary search bounds
    for _ in range(100):
        mid = (lo + hi) / 2
        if path_loss_UMa_NLOS(mid, fc_GHz) < Lmax_dB:
            lo = mid
        else:
            hi = mid
    return (lo + hi) / 2

Rmax_m  = find_rmax(Lmax, fc_GHz)  # = 843.3 m
Rmax_km = Rmax_m / 1000

# ── PART 2: CHANNEL MATRIX (3GPP TR 38.901 Section 7.5) ────────
fc_Hz  = 1.8e9;  c = 3e8;  lam = c / fc_Hz  # 0.1667 m
N_BS   = 4;  N_UE = 2           # MIMO dimensions
N_cl   = 20;  M_ray = 20        # clusters and rays/cluster
d_s    = lam / 2                # lambda/2 antenna spacing

# Step 2: Large Scale Parameters (Table 7.5-6 UMa-NLOS)
DS  = 10**np.random.normal(-6.955, 0.66)   # Delay Spread
ASA = 10**np.random.normal(1.81,  0.20)    # AoA Spread
ASD = 10**np.random.normal(1.59,  0.28)    # AoD Spread
ZSA = 10**np.random.normal(0.88,  0.31)    # ZoA Spread
ZSD = 10**np.random.normal(0.50,  0.35)    # ZoD Spread

# Step 3: Cluster delays (Eq 7.5-1)
r_tau = 2.3
X_n   = np.random.uniform(0, 1, N_cl)
tau_n = np.sort(-r_tau * DS * np.log(X_n))
tau_n -= tau_n.min()                # normalise to zero
```

```
# Step 4: Cluster powers (Eq 7.5-4)
Z_n     = np.random.normal(0, 3.0, N_cl)   # zeta = 3 dB
P_prime = np.exp(-tau_n*(r_tau-1)/(r_tau*DS)) * 10**(-Z_n/10)
P_n     = P_prime / P_prime.sum()     # normalise

# Step 5: Cluster angles (Eq 7.5-6a)
C_phi = 1.289;  C_theta = 1.178  # N_cl=20, Tables 7.5-2/3
phi_AOD = np.random.choice([-1,1],N_cl)*(2*ASD/1.4)*np.sqrt(-np.log(P_n/P_n.max()))/C_phi
phi_AOA = np.random.choice([-1,1],N_cl)*(2*ASA/1.4)*np.sqrt(-np.log(P_n/P_n.max()))/C_phi
theta_ZOD = np.random.choice([-1,1],N_cl)*ZSD*np.sqrt(-np.log(P_n/P_n.max()))/C_theta + 90
theta_ZOA = np.random.choice([-1,1],N_cl)*ZSA*np.sqrt(-np.log(P_n/P_n.max()))/C_theta + 90

# Step 6: Per-ray phases, XPR, and channel matrix construction
Phi    = np.random.uniform(-np.pi, np.pi, (N_cl, M_ray, 4))
kappa  = 10**(np.random.normal(7.0, 3.0, (N_cl, M_ray)) / 10)
alpha_m = np.array([0.0447,-0.0447,0.1413,-0.1413,0.2492,-0.2492,
                    0.3715,-0.3715,0.5129,-0.5129,0.6797,-0.6797,
                    0.8844,-0.8844,1.1267,-1.1267,1.4673,-1.4673,
                    1.8142,-1.8142])  # Table 7.5-5 ray offsets

def ULA(N, d_s, lam, phi_deg):
    n = np.arange(N)
    return np.exp(1j*2*np.pi*(d_s/lam)*n*np.sin(np.deg2rad(phi_deg)))

H = np.zeros((N_UE, N_BS), dtype=complex)
for n in range(N_cl):
    H_n = np.zeros((N_UE, N_BS), dtype=complex)
    for m in range(M_ray):
        phi_aod = phi_AOD[n] + ASD * alpha_m[m]
        phi_aoa = phi_AOA[n] + ASA * alpha_m[m]
        F_tx = ULA(N_BS, d_s, lam, phi_aod)
        F_rx = ULA(N_UE, d_s, lam, phi_aoa)
        pol  = np.exp(1j * Phi[n, m, 0])  # VV polarisation
        H_n += pol * np.outer(F_rx, F_tx)
    H += np.sqrt(P_n[n]) * H_n
H /= np.sqrt(M_ray)  # normalise by sqrt(M)

# SVD and capacity
U, S, Vh = np.linalg.svd(H)
print('Singular values:', S.round(4))
print('Rank:', np.sum(S > 1e-6))
SNR_lin  = 10**(3/10)  # 3 dB
capacity = sum(math.log2(1 + SNR_lin/N_BS * sv**2) for sv in S)
print(f'MIMO capacity: {capacity:.3f} bps/Hz')
```

5. References

[1] 3GPP TR 38.901 V18.0.0 (2024-05), "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project, May 2024.

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